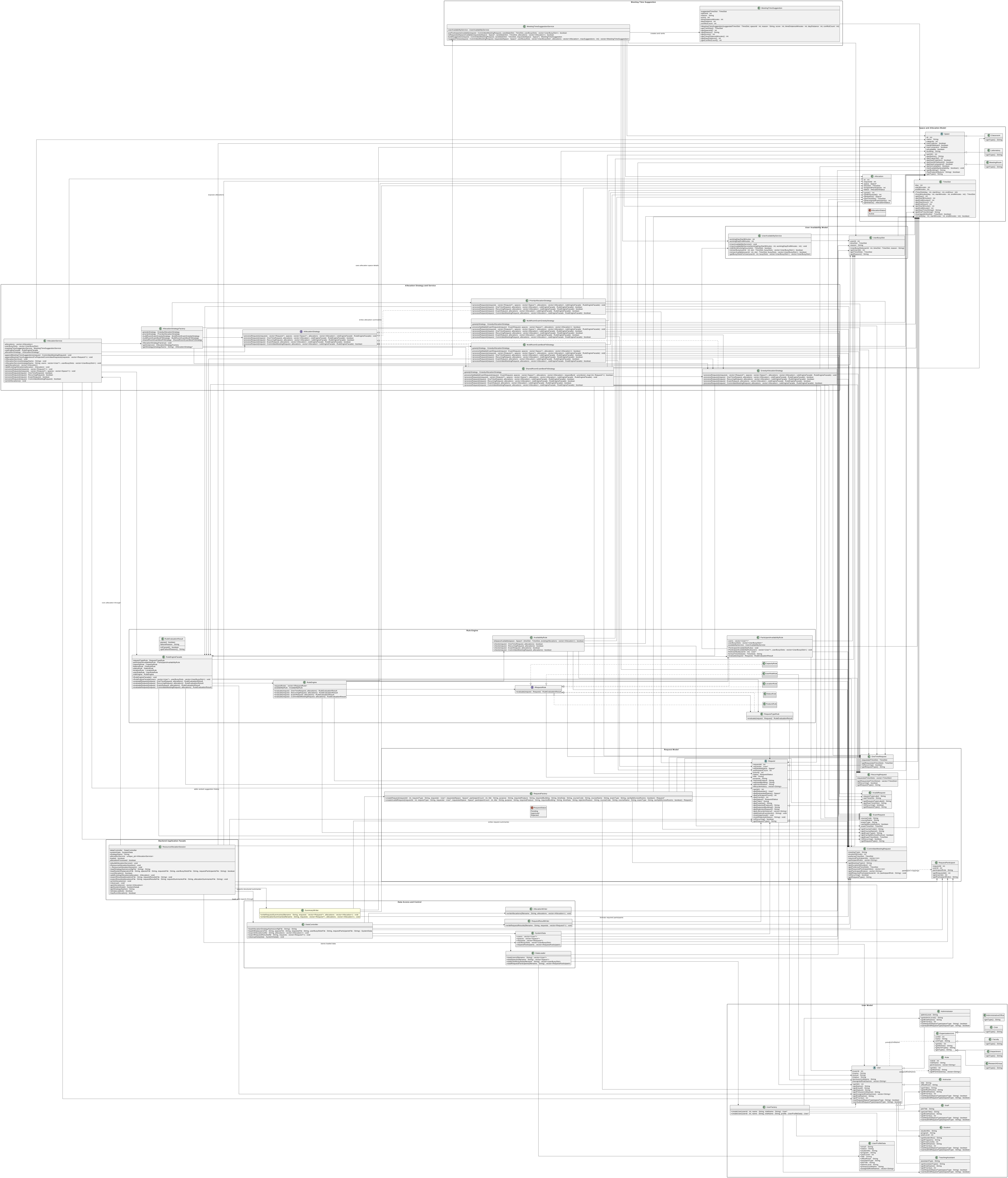
Meeting Time Suggestion
MeetingTimeSuggestionService
-userAvailabilityService : UserAvailabilityService
-areParticipantsAvailable(request : CommitteeMeetingRequest, candidateSlot : TimeSlot, userBusySlots : vector<UserBusySlot>) : boolean
-isRequestedSpaceAvailable(requestedSpace : Space*, candidateSlot : TimeSlot, allocations : vector<Allocation>) : boolean
-buildSuggestion(request : CommitteeMeetingRequest, candidateSlot : TimeSlot, requestedSpace : Space*) : MeetingTimeSuggestion
+suggestTimes(request : CommitteeMeetingRequest, requestedSpace : Space*, userBusySlots : vector<UserBusySlot>, allocations : vector<Allocation>, maxSuggestions : int) : vector<MeetingTimeSuggestion>
creates and ranks
MeetingTimeSuggestion
-suggestedTimeSlot : TimeSlot
-spaceId : int
-reason : String
-score : int
-timeDistanceMinutes : int
-dayDistance : int
-conflictCount : int
+MeetingTimeSuggestion(suggestedTimeSlot : TimeSlot, spaceId : int, reason : String, score : int, timeDistanceMinutes : int, dayDistance : int, conflictCount : int)
+getTimeSlot() : TimeSlot
+getSpaceId() : int
+getReason() : String
+getScore() : int
+getTimeDistanceMinutes() : int
+getDayDistance() : int
+getConflictCount() : int
Space and Allocation Model
Space
-id : int
-name : String
-capacity : int
-hasProjector : boolean
-hasWhiteboard : boolean
-hasComputers : boolean
-isAvailable : boolean
-building : String
+getId() : int
+getName() : String
+getCapacity() : int
+getHasProjector() : boolean
+getHasWhiteboard() : boolean
+getHasComputers() : boolean
+getIsAvailable() : boolean
+setAvailability(availability : boolean) : void
+getBuilding() : String
+hasFeature(feature : String) : boolean
+getType() : String
Classroom
+getType() : String
Laboratory
+getType() : String
MeetingRoom
+getType() : String
Allocation
-id : int
-requestId : int
-space : Space*
-timeSlot : TimeSlot
-assignedParticipants : int
-status : AllocationStatus
+getId() : int
+getRequestId() : int
+getSpace() : Space*
+getTimeSlot() : TimeSlot
+getAssignedParticipants() : int
+getStatus() : AllocationStatus
AllocationStatus
Active
TimeSlot
-day : int
-startMinutes : int
-endMinutes : int
+TimeSlot(day : int, startHour : int, endHour : int)
+fromMinutes(day : int, startMinutes : int, endMinutes : int) : TimeSlot
+getDay() : int
+getStartMinutes() : int
+getEndMinutes() : int
+getStartHour() : int
+getEndHour() : int
+getStartMinute() : int
+getEndMinute() : int
+getStartTimeString() : String
+getEndTimeString() : String
+overlapsWith(other : TimeSlot) : boolean
+isValid(day : int, startMinutes : int, endMinutes : int) : boolean
User Availability Model
UserAvailabilityService
-workingDayStartMinutes : int
-workingDayEndMinutes : int
+UserAvailabilityService() : void
+UserAvailabilityService(workingDayStartMinutes : int, workingDayEndMinutes : int) : void
+isWithinWorkingHours(slot : TimeSlot) : boolean
+isUserBusy(userId : int, slot : TimeSlot, busySlots : vector<UserBusySlot>) : boolean
+isUserAvailable(userId : int, slot : TimeSlot, busySlots : vector<UserBusySlot>) : boolean
+getBusySlotsForUser(userId : int, busySlots : vector<UserBusySlot>) : vector<UserBusySlot>
UserBusySlot
-userId : int
-timeSlot : TimeSlot
-reason : String
+UserBusySlot(userId : int, timeSlot : TimeSlot, reason : String)
+getUserId() : int
+getTimeSlot() : TimeSlot
+getReason() : String
exposes allocations
uses allocation space details
Allocation Strategy and Service
writes allocation summaries
AllocationService
-allocations : vector<Allocation>
-userBusySlots : vector<UserBusySlot>
-meetingTimeSuggestionService : MeetingTimeSuggestionService
-ruleEngineFacade : RuleEngineFacade
-allocationStrategy : IAllocationStrategy*
-appendMeetingTimeSuggestions(request : CommitteeMeetingRequest) : void
-appendMeetingTimeSuggestionsForRejectedCommitteeRequests(requests : vector<Request*>) : void
+AllocationService() : void
+AllocationService(strategyName : String) : void
+AllocationService(strategyName : String, users : vector<User*>, userBusySlots : vector<UserBusySlot>) : void
+getAllocations() : vector<Allocation>
+addExistingAllocation(allocation : Allocation) : void
+processRequests(requests : vector<Request*>) : void
+processRequests(requests : vector<Request*>, spaces : vector<Space*>) : void
+processRequest(request : OneTimeRequest) : boolean
+processRequest(request : RecurringRequest) : boolean
+processRequest(request : ExamRequest) : boolean
+processRequest(request : CommitteeMeetingRequest) : boolean
+printAllocations() : void
AllocationStrategyFactory
-greedyStrategy : GreedyAllocationStrategy
-priorityStrategy : PriorityAllocationStrategy
-multiRoomExamGreedyStrategy : MultiRoomExamGreedyStrategy
-multiRoomExamBestFitStrategy : MultiRoomExamBestFitStrategy
-sharedRoomExamBestFitStrategy : SharedRoomExamBestFitStrategy
-AllocationStrategyFactory() : void
+getInstance() : AllocationStrategyFactory
+getStrategy(strategyName : String) : IAllocationStrategy*
IAllocationStrategy
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
PriorityAllocationStrategy
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
MultiRoomExamGreedyStrategy
-greedyStrategy : GreedyAllocationStrategy
-processSplittableExamRequest(request : ExamRequest, spaces : vector<Space*>, allocations : vector<Allocation>) : boolean
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
MultiRoomExamBestFitStrategy
-greedyStrategy : GreedyAllocationStrategy
-processSplittableExamRequest(request : ExamRequest, spaces : vector<Space*>, allocations : vector<Allocation>) : boolean
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
SharedRoomExamBestFitStrategy
-greedyStrategy : GreedyAllocationStrategy
-processSplittableExamRequest(request : ExamRequest, spaces : vector<Space*>, allocations : vector<Allocation>, requestById : unordered_map<int, Request*>) : boolean
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
GreedyAllocationStrategy
+processRequests(requests : vector<Request*>, spaces : vector<Space*>, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : void
+processRequest(request : OneTimeRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : RecurringRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : ExamRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
+processRequest(request : CommitteeMeetingRequest, allocations : vector<Allocation>, ruleEngineFacade : RuleEngineFacade) : boolean
runs allocation through
Rule Engine
RuleEvaluationResult
-passed : boolean
-failureReason : String
+isPassed() : boolean
+getFailureReason() : String
RuleEngineFacade
-requestTypeRule : RequestTypeRule
-participantAvailabilityRule : ParticipantAvailabilityRule
-capacityRule : CapacityRule
-featureRule : FeatureRule
-statusRule : StatusRule
-locationRule : LocationRule
-userRoleRule : UserRoleRule
-ruleEngine : RuleEngine
+RuleEngineFacade() : void
+RuleEngineFacade(users : vector<User*>, userBusySlots : vector<UserBusySlot>) : void
+evaluateRequest(request : OneTimeRequest, allocations) : RuleEvaluationResult
+evaluateRequest(request : RecurringRequest, allocations) : RuleEvaluationResult
+evaluateRequest(request : ExamRequest, allocations) : RuleEvaluationResult
+evaluateRequest(request : CommitteeMeetingRequest, allocations) : RuleEvaluationResult
RuleEngine
-requestRules : vector<IRequestRule>
-availabilityRule : AvailabilityRule
+evaluate(request : OneTimeRequest, allocations) : RuleEvaluationResult
+evaluate(request : RecurringRequest, allocations) : RuleEvaluationResult
+evaluate(request : ExamRequest, allocations) : RuleEvaluationResult
+evaluate(request : CommitteeMeetingRequest, allocations) : RuleEvaluationResult
AvailabilityRule
-isSpaceAvailable(space : Space*, timeSlot : TimeSlot, existingAllocations : vector<Allocation>) : boolean
+check(request : OneTimeRequest, allocations) : boolean
+check(request : RecurringRequest, allocations) : boolean
+check(request : ExamRequest, allocations) : boolean
+check(request : CommitteeMeetingRequest, allocations) : boolean
ParticipantAvailabilityRule
-users : vector<User*>*
-userBusySlots : vector<UserBusySlot>*
-availabilityService : UserAvailabilityService
+ParticipantAvailabilityRule() : void
+ParticipantAvailabilityRule(users : vector<User*>, userBusySlots : vector<UserBusySlot>) : void
-findUserById(userId : int) : User*
-formatTimeSlot(slot : TimeSlot) : String
+evaluate(request : Request) : RuleEvaluationResult
CapacityRule
UserRoleRule
LocationRule
StatusRule
FeatureRule
IRequestRule
+evaluate(request : Request) : RuleEvaluationResult
RequestTypeRule
+evaluate(request : Request) : RuleEvaluationResult
Request Model
adds ranked suggestion history
RequestFactory
+createRequest(requestId : int, requestType : String, requester : User*, requestedSpace : Space*, participantCount : int, title : String, purpose : String, requiredFeature : String, requiredBuilding : String, timeData : String, courseCode : String, courseName : String, examType : String, canSplitAcrossRooms : boolean) : Request*
+createInvalidRequest(requestId : int, requestType : String, requester : User*, requestedSpace : Space*, participantCount : int, title : String, purpose : String, requiredFeature : String, requiredBuilding : String, timeData : String, rejectionReason : String, courseCode : String, courseName : String, examType : String, canSplitAcrossRooms : boolean) : Request*
RequestStatus
Pending
Approved
Rejected
writes request summaries
Request
-requestId : int
-requester : User*
-requestedSpace : Space*
-participantCount : int
-priority : int
-status : RequestStatus
-title : String
-purpose : String
-requiredFeature : String
-requiredBuilding : String
-rejectionReason : String
-lifecycleHistory : vector<String>
+getId() : int
+getRequester() : User*
+getRequestedSpace() : Space*
+getParticipantCount() : int
+getPriority() : int
+getStatus() : RequestStatus
+getTitle() : String
+getPurpose() : String
+getRequiredFeature() : String
+getRequiredBuilding() : String
+getRejectionReason() : String
+getLifecycleHistory() : vector<String>
+addHistoryEvent(event : String) : void
+markApproved() : void
+markRejected(reason : String) : void
+isRecurring() : boolean
+getRequestType() : String
OneTimeRequest
-requestedTimeSlot : TimeSlot
+getRequestedTimeSlot() : TimeSlot
+isRecurring() : boolean
+getRequestType() : String
RecurringRequest
-requestedTimeSlots : vector<TimeSlot>
+getRequestedTimeSlots() : vector<TimeSlot>
+isRecurring() : boolean
+getRequestType() : String
InvalidRequest
-requestTypeLabel : String
-rawTimeInfo : String
+getRequestTypeLabel() : String
+getRawTimeInfo() : String
+isRecurring() : boolean
+getRequestType() : String
ExamRequest
-courseCode : String
-courseName : String
-examType : String
-canSplitAcrossRooms : boolean
-examTimeSlot : TimeSlot
+getCourseCode() : String
+getCourseName() : String
+getExamType() : String
+getCanSplitAcrossRooms() : boolean
+getExamTimeSlot() : TimeSlot
+isRecurring() : boolean
+getRequestType() : String
CommitteeMeetingRequest
-meetingTopic : String
-durationMinutes : int
-preferredTimeSlot : TimeSlot
-requiredParticipantIds : vector<int>
-participantRoles : vector<String>
+getMeetingTopic() : String
+getDurationMinutes() : int
+getPreferredTimeSlot() : TimeSlot
+getRequiredParticipantIds() : vector<int>
+getParticipantRoles() : vector<String>
+addRequiredParticipant(userId : int, participantRole : String) : void
+isRecurring() : boolean
+getRequestType() : String
participant mappings
RequestParticipant
-requestId : int
-userId : int
-participantRole : String
+getRequestId() : int
+getUserId() : int
+getParticipantRole() : String
Backend Application Facade
ResourceAllocationSession
-dataController : DataController
-systemData : SystemData
-strategyName : String
-allocationService : unique_ptr<AllocationService>
-loaded : boolean
-allocationProcessed : boolean
-rebuildAllocationService() : void
+ResourceAllocationSession() : void
+~ResourceAllocationSession() : void
+loadStrategyName(configFile : String) : String
+loadSystemData(usersFile : String, spacesFile : String, requestsFile : String, userBusySlotsFile : String, requestParticipantsFile : String) : boolean
+runAllocation() : boolean
+addExistingAllocation(allocation : Allocation) : void
+exportResults(allocationsFile : String, requestResultsFile : String) : void
+exportResults(allocationsFile : String, requestResultsFile : String, requestSummariesFile : String, allocationSummariesFile : String) : void
+printAllocations() : void
+cleanup() : void
+getAllocations() : vector<Allocation>
+getSystemData() : SystemData&
+getStrategyName() : String
+isDataLoaded() : boolean
+hasRunAllocation() : boolean
exports structured summaries
loads and exports through
Data Access and Control
SummaryWriter
+writeRequestSummaries(filename : String, requests : vector<Request*>, allocations : vector<Allocation>) : void
+writeAllocationSummaries(filename : String, requests : vector<Request*>, allocations : vector<Allocation>) : void
DataController
+loadAllocationStrategyName(configFile : String) : String
+loadAllData(usersFile : String, spacesFile : String, requestsFile : String, userBusySlotsFile : String, requestParticipantsFile : String) : SystemData
+exportAllocations(file : String, allocations : vector<Allocation>) : void
+exportRequestResults(file : String, requests : vector<Request*>) : void
+cleanupData(data : SystemData&) : void
stores loaded data
AllocationWriter
+writeAllocations(filename : String, allocations : vector<Allocation>) : void
formats required participants
RequestResultWriter
+writeRequestResults(filename : String, requests : vector<Request*>) : void
SystemData
+users : vector<User*>
+spaces : vector<Space*>
+requests : vector<Request*>
+userBusySlots : vector<UserBusySlot>
+requestParticipants : vector<RequestParticipant>
DataLoader
+loadUsers(filename : String) : vector<User*>
+loadSpaces(filename : String) : vector<Space*>
+loadUserBusySlots(filename : String) : vector<UserBusySlot>
+loadRequestParticipants(filename : String) : vector<RequestParticipant>
User Model
Administrator
-adminLevel : String
+getAdminLevel() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean
AdministrativeOffice
+getType() : String
Club
+getType() : String
OrganizationUnit
-unitId : int
-name : String
-unitType : String
+getId() : int
+getName() : String
+getUnitType() : String
+getType() : String
Faculty
+getType() : String
Department
+getType() : String
ResearchGroup
+getType() : String
primaryUnitName
Role
-roleId : int
-roleName : String
-permissions : vector<String>
+getId() : int
+getName() : String
+getPermissions() : vector<String>
assignedRoleNames
User
#userId : int
#name : String
#email : String
#status : String
#primaryUnitName : String
#assignedRoleNames : vector<String>
+getId() : int
+getName() : String
+getEmail() : String
+getStatus() : String
+getPrimaryUnitName() : String
+getAssignedRoleNames() : vector<String>
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean
Instructor
-title : String
-officeRoom : String
+getTitle() : String
+getOfficeRoom() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean
UserFactory
+createUser(userId : int, name : String, roleName : String) : User*
+createUser(userId : int, name : String, roleName : String, profile : UserProfileData) : User*
Staff
-jobTitle : String
+getJobTitle() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean
Student
-studentNo : String
-program : String
-yearLevel : int
+getStudentNo() : String
+getProgram() : String
+getYearLevel() : int
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean
UserProfileData
+email : String
+status : String
+studentNo : String
+program : String
+yearLevel : int
+title : String
+officeRoom : String
+assistantType : String
+jobTitle : String
+adminLevel : String
+primaryUnitName : String
+assignedRoleNames : vector<String>
TeachingAssistant
-assistantType : String
+getAssistantType() : String
+getRoleName() : String
+getPriority() : int
+canRequestSpaceType(spaceType : String) : boolean
+canSubmitRequestType(requestType : String) : boolean